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Spectral Criticality in a Generalized Einstein-Universe Sector

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We investigate the one-loop effective action of a massless scalar field propagating on a one-parameter family of isotropic ultrastatic backgrounds. This family contains two exact nontrivial solutions of the Einstein field equations in $(n+1)$ dimensions: a scalar-flat ultrastatic vacuum and a constant-curvature configuration. The associated Klein–Gordon operator is analyzed in a Sturm–Liouville framework and its functional determinant is regularized using spectral zeta-function techniques.

We identify the constant-curvature solution as a distinguished spectral critical point at which the Liouville transformation becomes logarithmic and the radial eigenfunctions change from Bessel modes to plane waves in the transformed coordinate. This qualitative change induces a logarithmic non-analyticity in the one-loop effective action, reflected in the renormalized Casimir energy. By contrast, the scalar-flat branch exhibits a distinct infrared behavior of the regularized one-loop effective action. These results establish a direct connection between the scaling properties of the underlying geometry and the spectral structure of quantum vacuum fluctuations, providing a new example of geometry-induced spectral critical behavior.

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Introduction

Historically, zero-point energy due to vacuum polarization was first studied in electromagnetic waves propagating inside cavities, in the so-called Casimir phenomena. Distortions of the vacuum can be produced by boundary conditions or confining potentials in the presence of conductors (classical electromagnetism), as well as by background fields curving Minkowski spacetime (semiclassical gravity). For instance, examples in the literature [1] have shown phase transition behavior arising from boundary conditions which are parameter-dependent, such as quasi- and pseudo-periodic. However, the consequences of parameter-dependent backgrounds remain comparatively less explored. This raises the question of whether geometry itself can drive critical behavior. Such effects may become relevant to semiclassical gravity and short-distance physics, where Casimir phenomena must be distinguished from possible deviations from Newtonian gravity [2] and tests of extra-dimensions [3–5]. As well, phenomenological high-energy physics, such as QCD bag models [6,7] and their dual QED superconductor approximations [8], still investigates the implications of Casimir energy in confined regions of spacetime.

In this work, our objective is to expand some classical computations of the Casimir energy (taken from electromagnetism) and apply them to parameter-dependent backgrounds of semiclassical gravity. On the technical side, in both the electromagnetic and quantum versions, one has to sum over all the possible eigenfrequencies. Since this usually leads to a divergent Casimir energy, such quantity has to be renormalized by heat-kernel or zeta-function regularization techniques [9–19]. A rigorous mathematical calculation of this sort can be made in terms of spectral geometry [11–14]. In this formalism, the operator of interest, such as the Hamiltonian or the Klein-Gordon equation, is decomposed in its eigenfunctions. Thus, we benefit from a highly-symmetrical geometry.

On these grounds, the identification of backgrounds for which the Klein–Gordon operator admits a separable decomposition is therefore of particular interest for both semiclassical gravity and mathematical physics. For example, ultrastatic spacetimes provide a useful framework [9,10], as they reduce the dynamics of quantum fields to a spectral problem on the spatial geometry. Motivated by these considerations, we investigate a one-parameter family of isotropic and ultrastatic power-law backgrounds. Two distinguished members of this family correspond to exact solutions of the Einstein field equations: a scalar-flat ultrastatic vacuum and a constant-curvature configuration with uniform energy density. The full family is then used to investigate how the spectral properties of the Klein–Gordon operator evolve under the influence of gravity.

In Section 1 we derive the reduced field equation and analyze its exact solution branches. In Section 2 we quantize a massless scalar field propagating on these backgrounds. Finally, in Section 3 we compute the one-loop effective action by spectral zeta-function regularization, discuss the associated Casimir energy, and show that the constant-curvature solution defines a distinguished spectral critical point of the Klein–Gordon operator.

1. Geometry and Matter Content

In this section, we construct a class of ultrastatic, spherically symmetric geometries in $(n+1)$ dimensions and analyze the matter content required to support them. Higher-dimensional spacetimes have been extensively studied in classical gravity and provide a broad setting for investigations in semiclassical gravity [20]. We show that the Einstein field equations reduce to a single nonlinear, radial and second order differential equation for the metric potential, enabling a systematic characterization of stress-supported configurations. In particular, both the vacuum and uniform energy density solutions emerge as members of a broader power-law family of geometries exhibiting a conical structure and, depending on the parameters, naked singularities. These features make the resulting background a suitable setting for studying quantum field

propagation in the presence of nontrivial geometric defects while retaining analytical control over the spectral decomposition.

(a) Generalized Einstein-universe sector

We consider a generalized Einstein-universe sector described by the isotropic ultrastatic metric

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \Phi(r)\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j\,, \tag{1.1}$$

in $n+1$ spacetime dimensions. This class contains the Einstein static universe as a particular member when expressed in stereographic coordinates and extends it to a broader family of anisotropic stress-supported geometries.

Einstein field equations

The structure of the Einstein tensor suggests that the corresponding matter source must include anisotropic stresses. We therefore consider the most general static stress-energy tensor compatible with spherical symmetry, decomposed into energy density, isotropic pressure, and a traceless anisotropic component.

From Eqs. (A 6) of the Appendix A, the Einstein tensor components are

$$G_{00} = \frac{1}{2}R\,, \tag{1.2a}$$

$$G_{0i} = 0\,, \tag{1.2b}$$

$$G_{ij} = \left(\mathcal{A} - \frac{1}{2}\Phi R\right)\delta_{ij} + \mathcal{B}\frac{x^i x^j}{r^2}\,. \tag{1.2c}$$

Then, the Einstein field equations are satisfied by a static anisotropic stressenergy tensor of the form

$$T_{\mu\nu} = \rho(r)\,u_\mu u_\nu + P(r)\,h_{\mu\nu} + \tau_{\mu\nu}(r)\,, \tag{1.3}$$

where $u^\mu = \delta^\mu_0$ is the static unit timelike vector,

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu\,,$$

is the spatial projector, and $\tau_{\mu\nu}$ denotes a traceless anisotropic stress tensor.

The non–vanishing components are

$$\rho(r) = \frac{M_{\mathrm{P}}^2}{2}\,R\,, \tag{1.4a}$$

$$P(r) = \left(\frac{2-n}{n}\right)\frac{M_{\mathrm{P}}^2}{2}\,R\,, \tag{1.4b}$$

$$\tau_{ij}(r) = \Pi(r)\left(\frac{x^i x^j}{r^2} - \frac{1}{n}\delta_{ij}\right), \quad \Pi(r) = M_{\mathrm{P}}^2\mathcal{B}\,, \tag{1.4c}$$

$$\tau_{0\mu} = 0\,. \tag{1.4d}$$

Furthermore, from Eq. (1.4*b*) we see that the pressure is proportional to the energy density. Thus, the conservation law, $\nabla_\mu T^{\mu k} = 0$, yields the radial hydrostatic equation

$$P' + \frac{\Pi'}{\Phi} + \left[\frac{n-1}{r\Phi} + \frac{(n-3)\Phi'}{2\Phi^2}\right]\Pi = 0\,, \tag{1.5a}$$

$$P(r) = -\left(\frac{n-2}{n}\right)\rho(r)\,, \tag{1.5b}$$

which provides a consistency relation between the anisotropic stress $\Pi(r)$, the metric potential $\Phi(r)$, and the energy density $\rho(r)$.

Reduced field equation

For $n > 1$, the system of Eqs. (1.4) reduces to a non–linear second order differential equation for the metric potential, sourced by the energy density profile. We refer to this as the *reduced field equation*,

$$\frac{\Phi''}{\Phi} + \frac{n-1}{r}\frac{\Phi'}{\Phi} + \left(\frac{n-6}{4}\right)\left(\frac{\Phi'}{\Phi}\right)^2 + f(r)\Phi(r) = 0\,, \tag{1.6a}$$

$$f(r) := \frac{2\,\rho(r)}{(n-1)\,M_{\rm P}^2}\,. \tag{1.6b}$$

Consequently, once the reduced equation is solved for $\Phi(r)$, the matter variables follow from Eqs. (1.4), while Eq. (1.5) is automatically satisfied as a consequence of the conservation law. Notably, because of the nonlinearity, a given energy density profile does not determine a unique solution for the metric potential and multiple solution branches may arise.

In the trivial case of $n = 1$, both the energy density and the isotropic pressure vanish. Moreover, the anisotropic stress $\Pi(r)$ also vanishes identically since directional asymmetries cannot be sustained in a single spatial dimension, reducing the scenario to a trivial vacuum.

(b) Uniform energy density sector

Einstein static universe

We first consider a uniform and positive energy density,

$$\rho(r) = \rho_0 > 0\,, \tag{1.7a}$$

$$A := \frac{2\rho_0}{(n-1)M_{\rm P}^2}\,. \tag{1.7b}$$

Motivated by the stereographic form of the Einstein static universe, we consider the ansatz

$$\Phi(r) = \left(a + r^\alpha\right)^\beta\,. \tag{1.8}$$

Then, the reduced field equation admits the unique nontrivial solution

$$\alpha = 2\,, \qquad \beta = -2\,, \quad A = 4an\,. \tag{1.9}$$

Choosing the overall normalization to recover the standard stereographic parametrization yields

$$^{(\rm Ein)}\Phi(r) = \frac{4a^4}{(a^2 + r^2)^2}\,, \tag{1.10}$$

which coincides with the metric potential of the Einstein static universe in isotropic stereographic coordinates (see Appendix A).

From the Ricci scalar, Appendix A, Eq. (A 12), this geometry satisfies the Einstein field equations with

$$^{(\mathrm{Ein})}\rho = \frac{M_{\mathrm{P}}^2}{2}\frac{n(n-1)}{a^2}, \tag{1.11a}$$

$$^{(\mathrm{Ein})}P = -\frac{M_{\mathrm{P}}^2}{2}\frac{(n-1)(n-2)}{a^2}, \tag{1.11b}$$

$$^{(\mathrm{Ein})}\Pi = 0, \tag{1.11c}$$

corresponding to a constant curvature spacetime with a vanishing anisotropic stress.

Power-law sector

To exhibit another family of exact solutions for the same matter content, $\rho = \rho_0$, we consider the power-law ansatz

$$\Phi(r) = a\, r^b\,. \tag{1.12}$$

For $\rho = 0$, direct integration of Eq. (1.6) shows that any exponent b is admissible in $n = 2$ spatial dimensions, whereas for $n > 2$ the field equations uniquely select the scalar-flat branch $b = -4$. Eqs. (1.4*c*),(1.5*b*)&(A 14) further imply $P = \Pi = 0$, establishing this branch as an exact *ultrastatic* vacuum solution.

For uniform non-vanishing energy density, $\rho = \rho_0$, the same ansatz instead yields $b = -2$. From Appendix A, Eqs. (A 12)-(A 13), both the Ricci and Kretschmann scalars show uniform curvature across the entire spacetime,

$$^{(\mathrm{Pow})}R = \frac{(n-1)(n-2)}{a}\,, \tag{1.13a}$$

$$^{(\mathrm{Pow})}K = \frac{4(n-1)(n-4)^2}{a^2}\,. \tag{1.13b}$$

In this solution branch the matter content differs from the ultrastatic vacuum and the Einstein universe, as the Eqs. (1.4) imply

$$^{(\mathrm{Pow})}\rho(r) = \frac{M_{\mathrm{P}}^2}{2}\frac{(n-1)(n-2)}{a}\,, \tag{1.14a}$$

$$^{(\mathrm{Pow})}P(r) = -\frac{M_{\mathrm{P}}^2}{2}\frac{(n-1)(n-2)^2}{an}\,, \tag{1.14b}$$

$$^{(\mathrm{Pow})}\Pi(r) = -\frac{M_{\mathrm{P}}^2}{2}\frac{(n-2)}{r^2}\,. \tag{1.14c}$$

Notably, for $n \geq 3$, generic values of b produce a non-vanishing curvature (Appendix A, Eq. (A 12)) and hence an effective matter content. Although they are not exact solution branches of the present construction, we include them as a one-parameter geometric deformation for spectral analysis.

Singularities

For $n \geq 3$, from Eq. (A 12), the power-law profile yields a Ricci scalar which scales as

$$^{(\mathrm{Pow})}R(r) \propto b(b+4)\, r^{-b-2}\,. \tag{1.15}$$

The curvature behavior is therefore controlled by the exponent $-(b+2)$ and the prefactor $b(b+4)$. For generic values of b, the geometry exhibits curvature singularities.

When $-2 < b < 0$, $R(r) \to \infty$ as $r \to 0$, indicating a curvature singularity at the origin. In this regime the ultrastatic geometry has no event horizon and the singularity is naked (geodesically incomplete and causally accessible; see Appendix A).

When $b < -2$ (and $b \neq -4$), $R(r) \to 0$ as $r \to 0$, so the origin appears regular at the Ricci level. However, $R(r) \to \infty$ as $r \to \infty$, shifting the singular behavior to spatial infinity.

The special case $(b = -4)$ must be treated separately. Although the Ricci scalar vanishes, higher-order curvature invariants should be examined to assess the regularity of the geometry. As computed in Appendix A, Eq. (A 13), the Kretschmann scalar behaves as

$$^{(\mathrm{Pow})}K \propto r^{-2(b+2)}\,.$$

Therefore, for $b = -4$, the vanishing of the Kretschmann scalar, as $r \to 0$, indicates the absence of a curvature singularity at the origin, while its growth reflects a large-radius increase of curvature.

Conical behavior

The spatial geometry induced by the power-law profile exhibits a nontrivial angular structure.

As shown in Appendix A, Eq. (A 21), the relation between the proper circumference of a circle and its proper radius remains linear but with a modified proportionality factor. The constant rescaling of the angular sector implies that the geometry is locally equivalent to a cone, with a deficit or excess angle

$$\delta = -\pi b\,, \quad b \neq -2\,. \tag{1.16}$$

The case $b = -2$ is exceptional. In this limit the proper radial distance becomes

$$l(r) = \sqrt{a}\,\ln r, \tag{1.17}$$

while the proper circumference remains constant,

$$C(r) = 2\pi\sqrt{a}. \tag{1.18}$$

Hence the ratio between circumference and proper radius is no longer constant, and the notion of a conical deficit (or excess) angle ceases to be well defined. Instead, the spatial geometry acquires a cylindrical structure in the logarithmic radial coordinate.

Therefore, for $b \neq -2$, the power-law sector describes a defect-like geometry with either a deficit angle ($b < 0$) or an angular excess ($b > 0$). This conical structure coexists with the curvature singularities discussed above, and should be understood as a local geometric property away from the singular region. This definition coincides with the standard characterization of conical geometries, where the deficit angle is determined by the ratio between circumference and proper radius (see, for example, Ref. [21]).

From the perspective of quantum field propagation, this defect-like structure is expected to modify the angular mode decomposition of the field, leading to nontrivial spectral effects.

2. Quantum Correlators

In this section, we follow the semiclassical paradigm to study how correlators are affected by an external gravitational field curving Minkowski spacetime. Following the standard approach of canonical quantization (Ref. [22]), as well as spectral techniques (Refs. [11,23]), we construct the massless Wightman two-point function. In particular, we analyze the power-law profiles.

Klein-Gordon equation

Let us study the case of a free massless scalar field υ. The Klein-Gordon (KG) equation is given by

$$\ddot{\upsilon} - \frac{1}{\Phi}\boldsymbol{\nabla}^2\upsilon - \frac{n-2}{2\Phi^2}(\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\upsilon) = 0\,. \tag{2.1}$$

The radial symmetry of the background in $n \geq 2$ allows for a separation of variables in Eq. (2.1). Let us take

$$\upsilon(t, r, \Omega) := T(t)S(r)Y(\Omega)\,. \tag{2.2}$$

This ansatz leads to the system of equations

$$\ddot{T} + \omega^2 T = 0\,, \tag{2.3a}$$

$$\Delta_{n-1} Y + \ell(\ell+n-2) Y = 0\,, \tag{2.3b}$$

$$S'' + \left(\frac{n-1}{r} + \frac{n-2}{2}\frac{\Phi'}{\Phi}\right) S' + \left(\omega^2 \Phi - \frac{\ell(\ell+n-2)}{r^2}\right) S = 0\,, \tag{2.3c}$$

where Δ_{n-1} is the Laplace-Beltrami operator. In three dimensions, we have

$$\Delta_2 = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\,. \tag{2.4}$$

Kernel construction

In order to write the Klein–Gordon radial Eq. (2.3*c*) as a proper Sturm–Liouville problem, we introduce the Liouville transformation

$$\mathrm{d}x := \sqrt{\Phi}\,\mathrm{d}r\,, \quad \chi_{\ell\omega}(x) := r^{\frac{n-1}{2}}\Phi^{\frac{n-1}{4}}(r)\,S_{\ell\omega}(r)\,. \tag{2.5}$$

Then, we can write Eq. (2.3*c*) as

$$\left[-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + U(x)\right]\chi_{\ell\omega}(x) = \omega^2\,\chi_{\ell\omega}(x)\,, \tag{2.6}$$

where the effective Schrödinger potential

$$U(x) = \frac{1}{\Phi}\left[\frac{1}{r^2}\left(\sigma - \frac{1}{2}\right)\left(\sigma + \frac{1}{2}\right) + \frac{(n-1)(n-2)}{4r}\frac{\Phi'}{\Phi} + \frac{n-1}{4}\frac{\Phi''}{\Phi} + \frac{(n-1)(n-7)}{16}\left(\frac{\Phi'}{\Phi}\right)^2\right],$$
$$\sigma := \ell - 1 + \frac{n}{2}\,. \tag{2.7}$$

implicitly assumes $r = r(x)$.

Canonical quantization is then performed on the canonical field χ, while physical observables can subsequently be expressed in terms of the original field v.

Canonical Quantization

The quantum field $\hat{\chi}$ can be canonically quantized by different representations. Using the spherical symmetry of the background, we expand the canonical field in partial waves

$$\hat{\chi}(t,\mathbf{x}) = \sum_{\ell,m}\hat{\chi}_{\ell m}(t,r)\,Y_{\ell m}(\hat{\mathbf{x}})\,. \tag{2.8}$$

Following the spectral method of Graham [23], the mode expansion for each radial channel consists of a continuum of scattering states and possible discrete bound states

$$\hat{\chi}_{\ell m}(t,\mathbf{x}) = \int_0^\infty \frac{dk}{\sqrt{\pi\omega}}\left(\chi_{\ell\omega}(k,r)\,e^{-i\omega t}a_{\ell m}(k) + \chi^*_{\ell\omega}(k,r)\,e^{i\omega t}a^\dagger_{\ell m}(k)\right)$$
$$+ \sum_j \frac{1}{\sqrt{2\omega_j}}\left(\chi_{\ell j}(r)\,e^{-i\omega_j t}a_{\ell m j} + \chi^*_{\ell j}(r)\,e^{i\omega_j t}a^\dagger_{\ell m j}\right), \tag{2.9a}$$

$$\left[a_{\ell m}(k), a^\dagger_{\ell' m'}(k')\right] = \delta_{\ell\ell'}\delta_{mm'}\delta(k-k')\,, \tag{2.9b}$$

$$\left[a_{\ell m j}, a^\dagger_{\ell' m' j'}\right] = \delta_{\ell\ell'}\delta_{mm'}\delta_{jj'}\,, \tag{2.9c}$$

where the functions $\chi_{\ell j} := \chi_{\ell\omega_j}$ and $\chi_{\ell\omega}$ satisfy Eq. (2.6).

Moreover, since we are working in the massless case, we have $\omega = k$. The completeness relation then implies

$$\int_0^\infty \chi^*_\ell(\omega, x)\, \chi_\ell(\omega', x)\, \mathrm{d}x = \frac{\pi}{2}\, \delta(\omega - \omega'), \tag{2.10a}$$

$$\int_0^\infty \chi^*_{\ell j}(x)\, \chi_{\ell j'}(x)\, \mathrm{d}x = \delta_{jj'}. \tag{2.10b}$$

3. Casimir Energy for power-law ultrastatic backgrounds

Since we have defined the observables of interest from the propagator in the Kernel mode representation, we can now fix the radial functions. In order to analyze both the singular and conical regimes, we consider the power-law metric potential, Eq. (1.12), with $a > 0$, allowing b to vary to include configurations with curvature singularities as well as those exhibiting a purely conical profile. In this background (1.12) the radial KG equation (2.3c) can be written as

$$S''_\ell + \frac{1}{r}\left[(n-1) + \frac{b(n-2)}{2}\right] S'_\ell + \left(\omega^2 a r^b - \frac{\ell(\ell+n-2)}{r^2}\right) S_\ell = 0\,, \tag{3.1}$$

By the change of variable, for $b \neq -2$,

$$S(\rho) = \rho^{\frac{2-n}{2}}\, Z(\rho)\,, \quad \rho := \frac{2\sqrt{a}\omega}{b+2} r^{\frac{b+2}{2}}\,, \tag{3.2}$$

Eq. (3.1) can be transform into a Bessel equation

$$\frac{\mathrm{d}^2}{\mathrm{d}\rho^2} Z + \frac{1}{\rho}\frac{\mathrm{d}}{\mathrm{d}\rho} Z + \left(1 - \frac{\nu^2}{\rho^2}\right) Z = 0\,, \tag{3.3a}$$

$$\nu^2(\ell) := \beta + \frac{(n-2)^2}{4}\,, \quad \beta := \frac{4\ell(\ell+n-2)}{(b+2)^2}\,. \tag{3.3b}$$

The general solution is given by a linear combination of Bessel functions or Modified Bessel functions depending of the signs of a and ω. Since, b also changes the background geometry the choice of mode solutions should be studied carefully in accordance with the boundary conditions imposed. In particular, for $b > -2$, only the Bessel function of first kind is regular as

$$r \to 0\,, \quad J_\nu(\rho) \sim \rho^\nu\,.$$

Therefore, the physical solution is given by

$$S_{\ell\omega}(r) = A(\omega)\rho^{\frac{2-n}{2}} J_\nu(\rho)\,. \tag{3.4}$$

In terms of the Sturm–Liouville transformation, Eqs. (2.5)&(2.10), the normalized modes are

$$\chi_{\ell\omega}(x) = \sqrt{\rho\frac{\pi}{2}} J_\nu(\rho)\,, \quad x(\rho) = \frac{\rho}{\omega}\,, \quad x(r=0) = 0\,. \tag{3.5}$$

In contrast, for $b = -4$ the argument of the Bessel function exhibits an inverted radial scaling, leading to oscillatory behavior as $r \to 0$ (i.e. $\rho \to \infty$). However, in order to maintain a uniform comparison across all values of b, we consistently select the $J_\nu(\rho)$ branch of the Bessel solutions. This choice allows comparison with the Minkowski limit $b = 0$, with $a = 1$ fixing the normalization.

The case $b = -2$, which involves a qualitatively different Sturm–Liouville structure, is treated separately after the spherical zeta-function regularization is established.

(a) Spherical zeta function regularization

From the seminal work of [24], their regularized Casimir energy for Dirichlet internal modes is particularly useful for our power-law backgrounds. We can use their result almost verbatim for the one-loop calculation of the Casimir energy.

In this scenario, the quantization of the KG operator arise from boundary conditions. If the radial field is in a confined region, imposing Dirichlet

$$S_{\ell\omega}(L) = 0\,, \quad 0 \le r \le L\,,$$

gives

$$J_\nu\left(\frac{2\omega}{b+2}L^{\frac{b+2}{2}}\right) = 0\,.$$

The positive eigenfrequencies $\omega_{k\ell}$ for the spectral problem of Eq. (2.6) are given by

$$\omega_{k\ell} = \left|\frac{b+2}{2}\right| \frac{j_{\nu,k}}{L^{b/2+1}}\,, \tag{3.6}$$

where $j_{\nu,k}$ denotes the k-th positive zero of the Bessel function.

For $n > 1$, the angular sector possesses a non-trivial geometric multiplicity. We can cleanly isolate this by first defining the *partial-wave* zeta function for a fixed angular mode ν as

$$\zeta_\nu^{\mathcal{D}}(s) := \sum_{k=1}^{\infty} \left(j_{\nu,k}\right)^{-s}\,. \tag{3.7}$$

The *complete* spherical zeta function for the Dirichlet power-law n-sphere, denoted as $\mathrm{P}(\mathbb{S}^n)$, is constructed by summing over all partial waves weighted by their respective degeneracy degrees

$$\zeta^{\mathcal{D}}_{\mathrm{P}(\mathbb{S}^n)}(s;b) := \left|\frac{2L^{b/2+1}}{b+2}\right|^s \sum_{\ell=0}^{\infty} g_\ell\, \zeta_\nu^{\mathcal{D}}(s)\,, \tag{3.8a}$$

$$g_\ell = \frac{(2\ell+n-2)(\ell+n-3)!}{\ell!(n-2)!}\,. \tag{3.8b}$$

By substituting the generalized scale relation into the internal Dirichlet scalar 3D formula (Eq. 3.32 of [24]), the total spatial zeta function near $s=-1$ for an arbitrary parameter value $b \neq -2$ is given by

$$\zeta^{\mathcal{D}}_{\mathrm{P}(\mathbb{S}^3)}(s;b) = \left|\frac{b+2}{2L^{b/2+1}}\right| \left[\frac{2}{315\pi}\left(\frac{1}{s+1} - \ln\left|\frac{b+2}{2}\right| + \left(\frac{b}{2}+1\right)\ln L\right) + 0.00889 + \mathcal{O}(s+1)\right]\,. \tag{3.9}$$

The Casimir energy is defined via zeta-function regularization as

$$E_{\mathrm{Cas}}^{\mathrm{P}(\mathbb{S}^3)}(b) := \frac{1}{2}\,\mathrm{FP}_{s=-1}\,\zeta^{\mathcal{D}}_{\mathrm{P}(\mathbb{S}^3)}(s;b)\,, \tag{3.10}$$

where the finite part is taken after analytic continuation. Therefore,

$$\boxed{E_{\mathrm{Cas}}^{\mathrm{P}(\mathbb{S}^3)}(b) = \left|\frac{b+2}{4L^{b/2+1}}\right| \left[0.00889 + \frac{b+2}{315\pi}\ln L - \frac{2}{315\pi}\ln\left|\frac{b+2}{2}\right|\right], \qquad b \neq -2.} \tag{3.11}$$

(b) Physical interpretation

The renormalized Casimir energy $E_{\mathrm{Cas}}(b)$, Fig. 1, exhibits a non-analytic behavior at $b=-2$, where its first derivative with respect to the deformation parameter becomes discontinuous. This signals a qualitative change in the spectral dependence of the vacuum energy on the underlying geometric deformation.

From the perspective of the radial eigenvalue problem, the parameter b controls the effective scaling of the Sturm–Liouville operator associated with the KG equation. For generic b, the radial

equation defines a scale-dependent effective potential whose spectral properties vary nontrivially with the deformation parameter.

The origin of the non-analytic behavior can be traced to the structure of the associated Sturm–Liouville problem, which is most transparent in Liouville coordinates.

Critical case $b = -2$ and logarithmic Sturm–Liouville structure

In Sturm–Liouville coordinates, the effective radial equation admits a universal scaling form for power-law deformations $\Phi(r) \sim r^b$, namely

$$U(r) = \frac{c}{r^{b+2}}\,, \tag{3.12a}$$

$$c := \sigma^2 - \frac{1}{4} + \frac{b(n-1)(n+b-3)}{4} + \frac{b^2(n-1)(n-7)}{16}\,. \tag{3.12b}$$

At $b = -2$, the effective Schrödinger equation (2.6) reduces to a harmonic oscillator in the coordinate x, and admits solutions of the form

$$\chi_{\ell\omega}(x) = C_1 \cos(\alpha x) + iC_2 \sin(\alpha x)\,, \tag{3.13a}$$

$$x = \ln r\,, \tag{3.13b}$$

$$\alpha := \sqrt{c - \omega^2}\,. \tag{3.13c}$$

Thus, the radial eigenfunctions become plane waves in the logarithmic coordinate, in contrast with the Bessel-type solutions, Eq. (3.5), for $b \neq -2$, where the radial dependence is governed by a power-law reparametrization. This change in functional form alters the spectral representation entering the zeta-regularized Casimir energy and underlies the non-analytic behavior at $b = -2$.

This is reflected in the analytic structure of the associated zeta function, which develops a logarithmic dependence on b near $b = -2$, yielding a non-analytic contribution of the form $(b + 2)\ln|b + 2|$ in the renormalized Casimir energy. Consequently, near $b = -2$ one finds

$$E_{\rm Cas}^{{\rm P}(\mathbb{S}^3)}(b) \sim (b+2)\ln|b+2| + \text{regular terms}, \tag{3.14}$$

from which the derivative structure follows,

$$\partial_b E_{\rm Cas}^{{\rm P}(\mathbb{S}^3)}(b) \sim \ln|b+2| + 1 + \cdots\,, \tag{3.15}$$

and

$$\partial_b^2 E_{\rm Cas}^{{\rm P}(\mathbb{S}^3)}(b) \sim \frac{1}{b+2}. \tag{3.16}$$

Geometrically, $b = -2$ corresponds to a distinguished point where curvature invariants such as the Ricci and Kretschmann scalars become constant. While this does not indicate any change in topology or global structure, it reflects a modification in the radial curvature profile and hence in how quantum fluctuations probe the geometry.

Overall, $b = -2$ defines a spectral critical point separating two regimes with distinct scaling properties of the eigenvalue problem. The non-analytic behavior of the one-loop effective action reflects this change in spectral scaling, without invoking thermodynamic or symmetry-breaking interpretations.

By contrast, the scalar-flat branch $b = -4$ exhibits a logarithmic growth of the regularized one-loop effective action with the infrared scale L. Whether this behavior originates from an intrinsic infrared enhancement of the underlying spectrum or requires an additional normalization of the effective action remains an open question and deserves further investigation.

4. Final discussion

In this work, we computed the zeta-regularized Casimir energy of a massless scalar field propagating on a one-parameter family of isotropic ultrastatic backgrounds that includes two

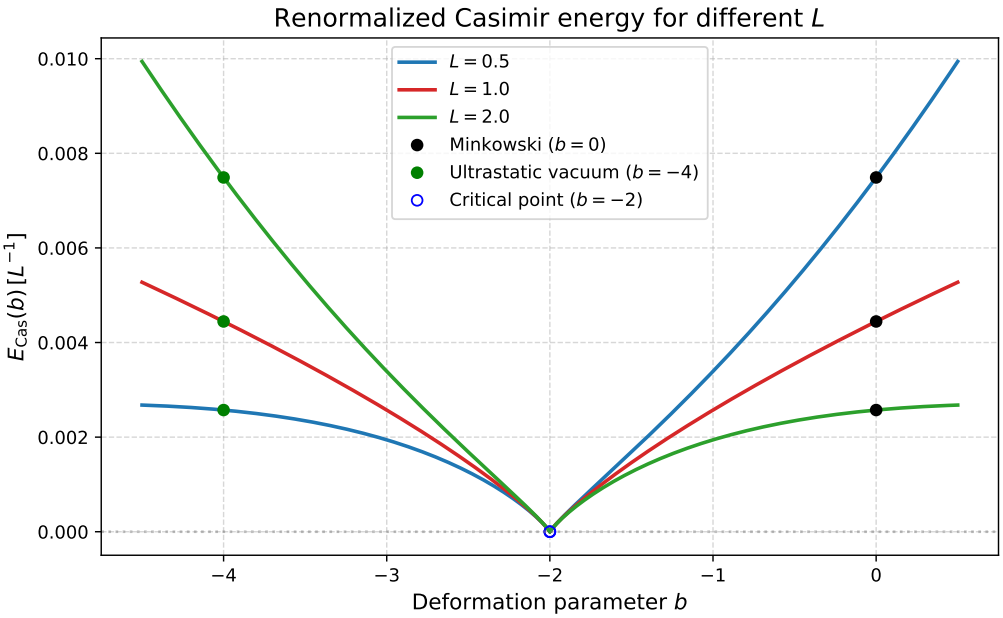


Figure 1: The renormalized Casimir energy $E_{\mathrm{Cas}}(b)$ for different values of the characteristic length scale L. The non-analytic behavior at the scale-invariant geometry, $b=-2$, is manifested by a singular derivative with respect to the deformation parameter, indicating a spectral critical point of the underlying Sturm–Liouville problem.

exact nontrivial solutions of the Einstein field equations in $(n+1)$ dimensions. The first corresponds to a scalar-flat ultrastatic vacuum, whereas the second describes a constant-curvature configuration with uniform energy density and isotropic pressure.

To this end, we first constructed the complete family of ultrastatic geometries compatible with the Einstein equations under the assumption of an isotropic metric potential. The corresponding matter sector is characterized by an isotropic equation of state relating the pressure and energy density, while more general anisotropic stress configurations are also admitted. This distinguishes the present family from previously studied ultrastatic geometries, such as the Einstein static universe, which is recovered as a particular case with vanishing anisotropic stress.

The main result of this work is the identification of the constant-curvature solution ($b=-2$) as a spectral critical point of the associated Klein–Gordon operator. At this value of the deformation parameter, the one-loop effective action acquires a logarithmic non-analytic dependence on the background geometry through the functional determinant of the Klein–Gordon operator, which is reflected in the renormalized Casimir energy. This behavior originates from a qualitative change in the spectral problem: the Liouville transformation becomes logarithmic, $x\sim\ln r$, and the radial eigenfunctions change from Bessel modes to plane waves in the logarithmic coordinate. In contrast, the scalar-flat ultrastatic vacuum ($b=-4$), the Minkowski limit ($b=0$), and the remaining power-law backgrounds are all described by Bessel-type solutions in monomial coordinates $x\sim r^{b/2+1}$.

Future work may proceed in several directions. The two exceptional branches identified in this work deserve a dedicated spectral analysis. For the critical geometry $(b=-2)$, whose logarithmic Liouville transformation leads to plane-wave eigenfunctions, it remains to determine whether a natural self-adjoint extension and boundary prescription can be defined, and whether a corresponding zeta-regularized one-loop effective action exists. Equally important is the scalar-flat branch $(b=-4)$, whose regularized one-loop effective action exhibits a logarithmic infrared growth. It would be worthwhile to clarify whether this behavior originates from an intrinsic

infrared enhancement of the spectrum, the endpoint structure of the associated Sturm–Liouville operator, or an additional normalization of the effective action.

A. Curvature calculations

The only non–vanishing Christoffel symbols of the metric ansatz (1.1) are

$$\Gamma^{i}{}_{jk} = \frac{1}{2}\left(\delta^{i}{}_{k}\frac{\partial_j\Phi}{\Phi} + \delta^{i}{}_{j}\frac{\partial_k\Phi}{\Phi} - \delta_{jk}\delta^{il}\frac{\partial_l\Phi}{\Phi}\right). \tag{A 1}$$

Eqs. (A 1) implicate that the non–vanishing components of the Riemann tensor are

$$R^{i}{}_{jkl} = \frac{1}{2\Phi^2}\left[\left(\delta^i_l\delta_{jk} - \delta^i_k\delta_{jl}\right)\boldsymbol{\nabla}\Phi\cdot\boldsymbol{\nabla}\Phi + 2\left(\delta^i_k\Phi_l - \delta^i_l\Phi_k\right)\Phi_j + 2\left(\delta_{jl}\Phi_k - \delta_{jk}\Phi_l\right)\Phi^{|i}\right]$$
$$+ \frac{1}{2\Phi}\left[\delta^i_l\Phi_{kj} - \delta^i_k\Phi_{lj} + \delta_{jk}\Phi^{|i}_{l} - \delta_{jl}\Phi^{|i}_{k}\right], \tag{A 2}$$

where $\Phi^{li} := \delta^{ij}\partial_j\Phi$ and $\Phi_j := \partial_j\Phi$. From this we can compute the Ricci tensor components as

$$R_{ij} = \left[-\frac{1}{2}\frac{\boldsymbol{\nabla}^2\Phi}{\Phi} + \left(1-\frac{n}{4}\right)\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^2}\right]\delta_{ij} + \left(1-\frac{n}{2}\right)\frac{\partial_i\partial_j\Phi}{\Phi} + \frac{3}{4}(n-2)\frac{\partial_i\Phi\partial_j\Phi}{\Phi^2}. \tag{A 3}$$

The Ricci scalar can be computed as

$$R = (1-n)\frac{\boldsymbol{\nabla}^2\Phi}{\Phi^2} - \frac{(n-1)(n-6)}{4}\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^3}. \tag{A 4}$$

On the other hand, the Kretschmann scalar is found to be

$$K = \frac{n(n-1)}{2}\frac{|\boldsymbol{\nabla}\Phi|^4}{\Phi^6} + 2(n-1)\frac{|\boldsymbol{\nabla}\Phi|^2}{\Phi^5}\boldsymbol{\nabla}^2\Phi + (n+1)\frac{(\boldsymbol{\nabla}^2\Phi)^2}{\Phi^4} - 2\frac{\Phi^{|ij}\Phi_{ij}}{\Phi^4}. \tag{A 5}$$

After imposing the isotropy of the metric potential we can write the Ricci spatial components in terms of radial functions as

$$R_{ij} = \mathcal{A}(r)\delta_{ij} + \mathcal{B}(r)\frac{x^i x^j}{r^2}, \tag{A 6a}$$

$$\mathcal{A} = -\frac{1}{2}\frac{\Phi''}{\Phi} + \left(\frac{3}{2}-n\right)\frac{1}{r}\frac{\Phi'}{\Phi} + \frac{1}{2}\left(2-\frac{n}{2}\right)\left(\frac{\Phi'}{\Phi}\right)^2, \tag{A 6b}$$

$$\mathcal{B} = \left(1-\frac{n}{2}\right)\frac{\Phi''}{\Phi} + \left(\frac{n}{2}-1\right)\frac{1}{r}\frac{\Phi'}{\Phi} + \frac{3}{4}(n-2)\left(\frac{\Phi'}{\Phi}\right)^2, \tag{A 6c}$$

$$f' := \frac{\mathrm{d}f}{\mathrm{d}r}, \quad r := |\mathbf{x}|.$$

Metric potential for the Einstein universe

To transform the spatial geometry of the Einstein static universe from standard hyperspherical to stereographic coordinates, we apply a stereographic projection. Starting with the spatial metric on an n-sphere of radius a,

$$\mathrm{d}l^2 = a^2(\mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega^2_{n-1}), \tag{A 7}$$

we introduce the stereographic radial coordinate
$r = a\tan(\chi/2)$.

Substituting the corresponding differential and identity

$$\mathrm{d}\chi = \frac{2a}{a^2+r^2}\,\mathrm{d}r\,, \quad \sin\chi = \frac{2ar}{a^2+r^2}\,, \tag{A 8}$$

into the line element yields

$$\mathrm{d}l^2 = \frac{4a^4}{(a^2+r^2)^2}(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_{n-1}^2)\,. \tag{A 9}$$

Recognizing that the term in parentheses is simply the flat Euclidean metric $\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j$ written in spherical coordinates, the full spacetime metric elegantly takes the manifestly conformally flat, isotropic form

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{4a^4}{(a^2+r^2)^2}\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j\,. \tag{A 10}$$

Power-law curvature

Let us compute the radial derivatives for $\Phi(r)$ of Eq. (1.12) in n dimensions

$$|\boldsymbol{\nabla}\Phi|^2 = a^2b^2r^{2b-2}\,, \tag{A 11a}$$

$$\boldsymbol{\nabla}^2\Phi = ab(b+n-2)r^{b-2}\,. \tag{A 11b}$$

After substituting these into Eq. (A 4) for the Ricci scalar we have

$$R = -\frac{b(b+4)(n-1)(n-2)}{4ar^{b+2}}\,. \tag{A 12}$$

From Eq. (A 5), the Kretschmann scalar is given by

$$K = \frac{b^2(n-1)}{2a^2}\,r^{-2(b+2)}\left[(n+6)b^2 + 8(n-1)b + 2(n^2-2n-4)\right]\,. \tag{A 13}$$

The $\mathcal{B}$ is then

$$\mathcal{B} = \frac{(n-2)b(b+4)}{4r^2}\,. \tag{A 14}$$

Conical structure of the power-law sector

Using the identity

$$\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j = \mathrm{d}r^2 + r^2\mathrm{d}\Omega_{n-1}^2, \tag{A 15}$$

the spatial line element becomes

$$\mathrm{d}l^2 = \Phi(r)\left(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_{n-1}^2\right)\,. \tag{A 16}$$

The proper radial distance, obtained from the spatial line element in the standard way (see Ref. [25]), is

$$l(r) = \int_0^r \sqrt{\Phi(r')}\,\mathrm{d}r' = \sqrt{a}\int_0^r (r')^{b/2}\,\mathrm{d}r'\,, \tag{A 17}$$

where the angular coordinates are held fixed, $\mathrm{d}\Omega_{n-1} = 0$.

For $b \neq -2$, this yields

$$l(r) = \frac{\sqrt{a}}{1+\frac{b}{2}}\, r^{1+\frac{b}{2}}\,. \tag{A 18}$$

At fixed radius, the proper circumference is

$$C(r) = \int_0^{2\pi} \sqrt{\Phi(r)}\, r\, \mathrm{d}\theta = 2\pi\sqrt{a}\, r^{1+\frac{b}{2}}\,. \tag{A 19}$$

The ratio between circumference and proper radius is therefore

$$\alpha = \frac{C(r)}{2\pi\, l(r)} = 1 + \frac{b}{2}\,, \tag{A 20}$$

which is independent of r. This shows that the angular sector is uniformly rescaled, corresponding to a conical geometry with deficit (or excess) angle

$$\delta = 2\pi(1-\alpha) = -\pi b\,. \tag{A 21}$$

This establishes that the spatial geometry associated with the ultrastatic vacuum solution and the power-law sector is locally conical.

Geodesic incompleteness and nakedness

We analyze radial time-like geodesics in the ultrastatic background, Eq. (1.1). The conserved energy E implies

$$\dot{r}^2 = \frac{E^2-1}{\Phi(r)}\,. \tag{A 22}$$

The time required for an observer starting at radius r_0 to reach the origin is therefore

$$\Delta t = \int_0^{r_0} \sqrt{\Phi(r)}\, dr\,. \tag{A 23}$$

For the power-law configuration (1.12), the conformal factor behaves as

$$\sqrt{\Phi(r)} \sim r^{b/2}\,. \tag{A 24}$$

The integral converges whenever $b > -2$. In this regime, time-like geodesics reach $r = 0$ in finite proper time, establishing geodesic incompleteness. This condition coincides with the divergence of curvature invariants, $R(r) \propto r^{-(b+2)}$, confirming the presence of a physical singularity at the origin.

Due to the ultrastatic nature of the metric, one has $g_{tt} = -1$ everywhere, so no redshift factor develops and no event horizon can form. Consequently, signals from arbitrarily small radii can propagate to distant observers. Combining geodesic incompleteness with the absence of a horizon, we conclude that, for $-2 < b < 0$, the power-law configuration contains a naked singularity at $r = 0$.

Data accessibility

This article has no additional data.

Author's contributions

A.C.E. conceived the study, performed the analytical calculations, carried out the spectral analysis, and wrote the manuscript. The author approved the final version of the manuscript.

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The author declares no competing interests.

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